

⇒ Integration by parts in hindie

⇒  $U = f(n)$  ,  $V = g(n)$

$$\int U \cdot v \, dn = U \int v \, dn - \int \left[ \frac{dU}{dn} \cdot \int v \, dn \right] dn$$
$$\int U \cdot v \, dn = U \int v \, dn - \int \frac{dU}{dn} \cdot \int v \, dn \, dn$$

→ Using ILATE we can choose that which function we have to take as  $U$  or  $V$ .

$I$  = Inverse function,  $L$  = logarithm,  $A$  = Algebra  
 $P$  = Trigonometry,  $E$  = exponential

ex/  $\int n \cos n \, dn$

Using ILATE

Let  $U = n$  and  $V = \cos n$ .

$$\int U \cdot v = U \int v \, dn - \int \left[ \frac{dU}{dn} \right] \cdot \int v \, dn$$

$$n \int \cos n - \int \frac{dn}{dn} \cdot \int \cos n$$



$$n(-\cos n) - \int 1 \cdot (-\cos n)$$

$$n(-\cos n) + \sin n + c$$

$$\boxed{n - \cos n + \sin n + c} \text{ Ans}$$